

Solution Requirements

Date	25 June 2026
Project Name	Smart Lender
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the loan prediction page without logging in. (If login is implemented later, secure authentication will be used.)	MEDIUM
2	Authorization levels	General users can enter loan details and receive predictions. The administrator can update the ML model and maintain the application.	MEDIUM
3	External interfaces	The system interacts with the trained XGBoost model, scaler file, and Flask web application for predictions.	HIGH

4	Transactions processing	The system accepts user input, preprocesses the data, predicts loan eligibility using the ML model, and displays the result instantly.	HIGH
5	Reporting	The application displays the loan prediction result on the webpage. Future versions may allow downloading reports.	MEDIUM
6	Business rules, etc.	Loan eligibility is determined based on applicant details such as income, education, employment status, credit history, loan amount, and other input features using the trained ML model.	HIGH
7	Compliance to laws or regulations	User information is processed securely and used only for prediction purposes while maintaining data privacy.	MEDIUM
8	<i>Other</i>	The application validates user inputs before making predictions to reduce errors.	MEDIUM

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate predictions quickly after the user submits the form.	Prediction displayed in less than 2 seconds
2	Scalability	The application should support multiple users without significant performance degradation.	Supports 100+ concurrent users
3	Security & Data Privacy	User input should be securely processed, and sensitive information should not be permanently stored.	Secure processing with input validation
4	Reliability & Availability	The application should remain available and provide consistent prediction results.	99% uptime during operation
5	Usability & Accessibility	The interface should be simple, user-friendly, and responsive on both desktop and mobile devices.	Easy navigation and responsive design
6	<i>Other</i>	The ML model and Flask application should be easy to update and maintain without affecting system functionality.	Easy model replacement and code maintenance